

GEORGIA INSTITUTE OF TECHNOLOGY
COE3001-H MECHANICS OF DEFORMABLE BODIES

Final exam - 2025 Fall

Dec. 11, 8:00-10:50 am

Name: _____ GTID: _____

Please read the following instructions

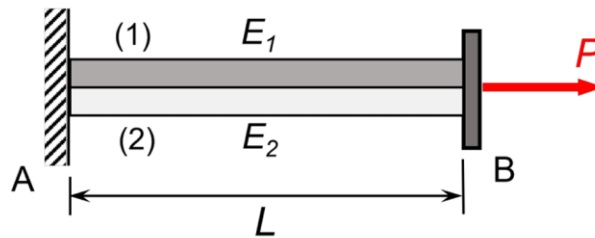
- ✧ You have **170** minutes to complete this exam.
- ✧ You may use a single, double-sided, letter sized formula sheet.
- ✧ This question booklet contains **8** questions, **18** pages (including the cover) for a total of **100** points. Check to see if any pages are missing.
- ✧ You are allowed to tear off **Page 18**, which does not need to be submitted. Page 1-17 cannot be damaged.
- ✧ DO NOT scribble or do rough work or make any stray marks on it. Use separate sheet for rough work.
- ✧ All the questions are compulsory, and all the notations have their usual meaning.
- ✧ Show all necessary steps for full credit. Clearly indicate your final answers.

Problem 1 (5 points)

A rod is made up of elements 1 and 2, each having a length of L and cross-sectional area A . Element 1 has an elastic modulus of E_1 , and element 2 has a modulus of E_2 , with $E_2 > E_1$. Let F_1 and F_2 represent the axial load carried by elements 1 and 2, respectively.

Choose the correct answer below:

- a) $|F_1| > |F_2|$
- b) $|F_1| = |F_2|$
- c) $|F_1| < |F_2|$
- d) More information is needed in order to answer this question.

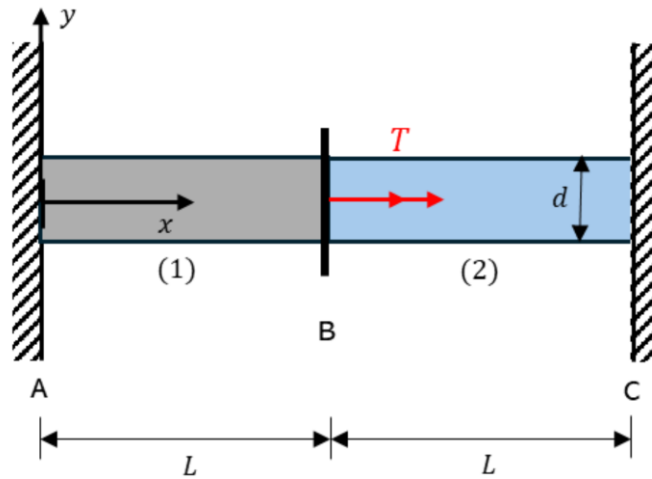


(If you are not sure whether your answer is correct, write your analysis below to get partial credit.)

Problem 2 (5 points)

A shaft consists of two segments of different materials ($G_1 = 3G_2$) but equal length and diameter, connected by a rigid plate at point B . If a torque T is applied at B , the magnitude of the reaction torques at the fixed ends A and C are:

- a) $|T_A| > |T_C|$
- b) $|T_A| = |T_C|$
- c) $|T_A| < |T_C|$
- d) More information is needed in order to answer this question.

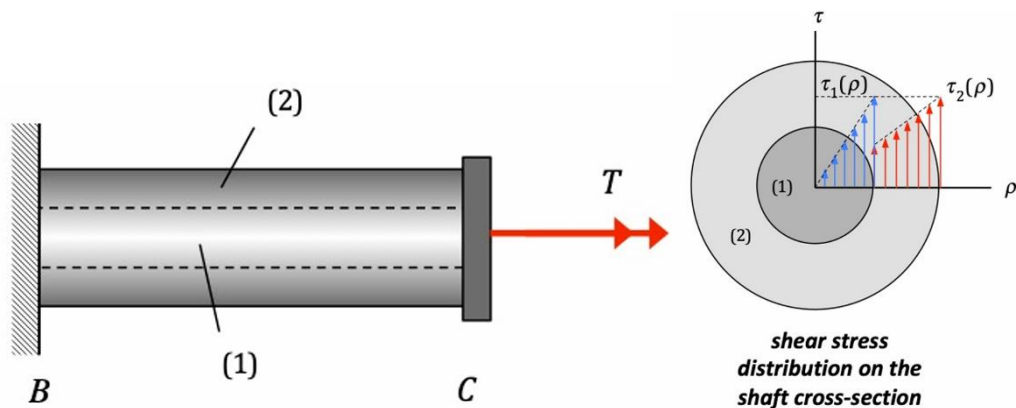


(If you are not sure whether your answer is correct, write your analysis below to get partial credit.)

Problem 3 (5 points)

A shaft is made of a shell (2) and core (1) with an axial torque T applied at the connector C. There is **no relative slide** between shell and core. Let G_1 and G_2 represent the shear modulus of (1) and (2), respectively. The shear stress distribution across the shaft cross-section is shown in the figure. Note from the figure that the maximum shear stresses in (1) and (2) are equal. Choose the correct answer below:

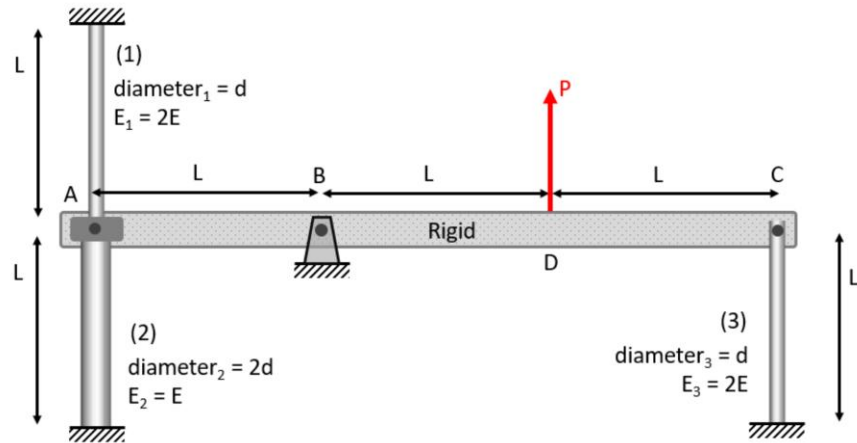
- a) $G_1 > G_2$
- b) $G_1 = G_2$
- c) $G_1 < G_2$
- d) More information is needed in order to answer this question.



(If you are not sure whether your answer is correct, write your analysis below to get partial credit.)

Problem 4 (15 points)

Rigid bar ABC is held by a pin joint at B and connected to three deformable rods. Two deformable rods are connected at A, and one deformable rod is connected at C. A force P is applied at point D. The diameters and elastic modulus of the deformable rods are listed in the figure.



- (a) Determine whether the assembly is determinate or indeterminate. (3 points)
- (b) What are the axial stresses in each of the deformable members? Leave your answers in terms of P , L , E , and/or d . (9 points)
- (c) If the allowable stress σ_a is 200 MPa, what is the maximum value of P that can be applied? Use values of $L = 1$ m, $E = 70$ GPa, and $d = 0.02$ m (3 points)

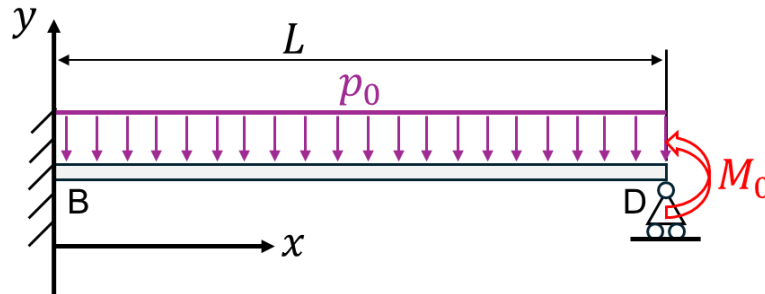
Assume that the deformation is small and the rods only deform in their axial direction.

Problem 5 (25 points)

The beam BD has a distributed load p_0 acting between B and D and a concentrated moment M_0 applied at D as shown. The beam is fixed at B and is supported by a roller at D. E and I are constant along the beam. Given

$$p_0 = \frac{16P}{L}, \quad M_0 = 2PL,$$

Use the second-order integration method to answer the following:



- (1) Draw the FBD for the beam BD. (3 pts)

- (2) Write the equilibrium equations for the beam BD and determine the degree of static indeterminacy. (5 pts)

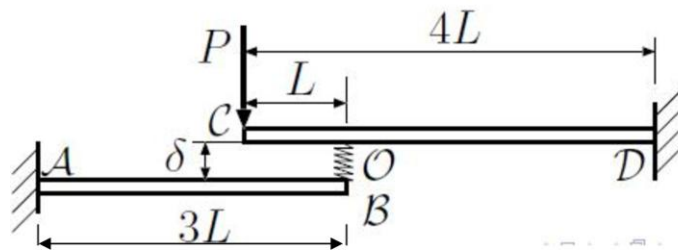
(3) What is the compatibility equation(s)? (2 pts)

(4) Calculate the reactions on the beam at B and D in terms of P and L by converting the compatibility equation(s) into equation(s) of unknowns using method of successive integrations. (10 pts)

- (5) Find the equation of the deflection curve, $v(x)$ throughout the beam in terms of P , L , E , and I . (5 pts)

Problem 6 (15 points)

As shown below, two prismatic cantilever beams AB and CD, both having Young's modulus E and moment of inertia I , are fixed at A and D, respectively. On beam AB, a spring with a constant k is fixed to the free end B (the spring is assumed to deform only in the vertical direction). Initially with no system loading, the spring has a length of δ ($\delta \ll L$) and its tip touches the beam CD at O with no force. Now a vertical force P is applied downward at C on beam CD. Determine the magnitude of P , such that at equilibrium, beam CD begins to make contact with beam AB by its free end C. Assume all contacts occurred are smooth and the horizontal displacement can be neglected. Also, for simplicity, take $EI = 9kL^3$ and express the result in terms of k and δ .

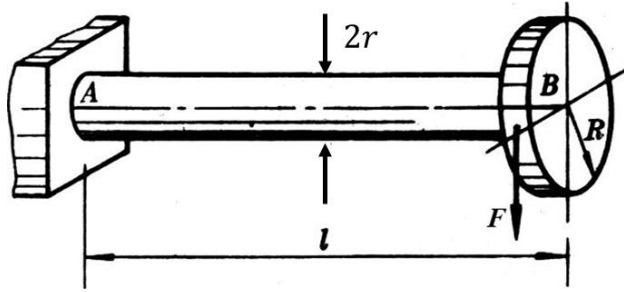


Problem 7 (15 points)

As shown below, for a solid circular shaft AB, one end A is fixed, and the free end B is assembled with a wheel. The external load F is applied on the edge of wheel. The length of the bar AB is l , the radius of the wheel is R , and the radius of the shaft is r .

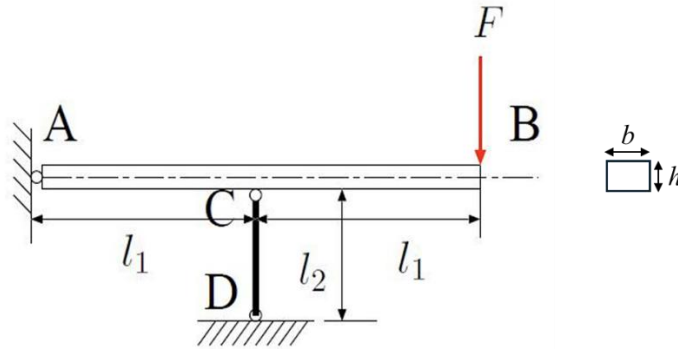
Determine the most dangerous point(s) according to the Maximum shear stress theory

(Hint: $\sigma_{r3} = \sigma_1 - \sigma_3$). What is the maximum principal stress at the dangerous point(s)?



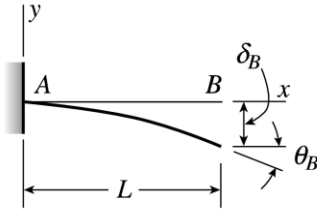
Problem 8 (15 points)

A horizontal beam with rectangular cross-section AB is pinned to a vertical wall at point A and loaded by a concentrated downward force F at its right end B. The beam is additionally supported at point C by a vertical solid circular rod CD, which is pinned at both end C and D, as shown in the figure. Given $l_1 = 1.25$ m, $l_2 = 0.55$ m, $b = 15$ cm, $h = 10$ cm. The allowable stress in AB is $[\sigma] = 180$ MPa. For the rod CD, the diameter $d = 25$ mm, Young's modulus $E = 200$ GPa, the proportional limit $\sigma_{pl} = 200$ MPa. The linear relation of inelastic buckling has coefficient $a = 310$ MPa and $b = 1.14$ MPa. The safety factor of stability is $n_{st} = 2$. Please determine the maximum allowable load F .



Deflection table

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Notation:

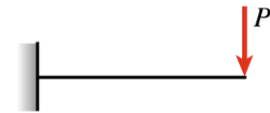
v = deflection in the y direction (positive upward)

$v' = dv/dx$ = slope of the deflection curve

$\delta_B = -v(L)$ = deflection at end B of the beam (positive downward)

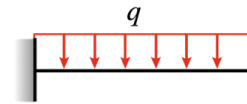
$\theta_B = -v'(L)$ = angle of rotation at end B of the beam (positive clockwise)

EI = constant



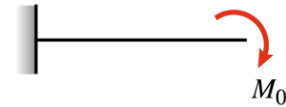
$$v = -\frac{Px^2}{6EI}(3L - x) \quad v' = -\frac{Px}{2EI}(2L - x)$$

$$\delta_B = \frac{PL^3}{3EI} \quad \theta_B = \frac{PL^2}{2EI}$$



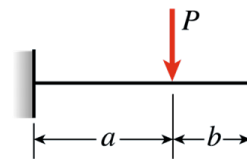
$$v = -\frac{qx^2}{24EI}(6L^2 - 4Lx + x^2) \quad v' = \frac{qx}{6EI}(3L^2 - 3Lx + x^2)$$

$$\delta_B = \frac{qL^4}{8EI} \quad \theta_B = \frac{qL^3}{6EI}$$



$$v = -\frac{M_0x^2}{2EI} \quad v' = -\frac{M_0x}{EI}$$

$$\delta_B = \frac{M_0L^2}{2EI} \quad \theta_B = \frac{M_0L}{EI}$$



$$v = -\frac{Px^2}{6EI}(3a - x) \quad v' = -\frac{Px}{2EI}(2a - x) \quad (0 \leq x \leq a)$$

$$v = -\frac{Pa^2}{6EI}(3x - a) \quad v' = -\frac{Pa^2}{2EI} \quad (a \leq x \leq L)$$

$$\text{At } x = a: \quad v = -\frac{Pa^3}{3EI} \quad v' = -\frac{Pa^2}{2EI}$$

$$\delta_B = \frac{Pa^2}{6EI}(3L - a) \quad \theta_B = \frac{Pa^2}{2EI}$$